

Software Engineering

Software Quality

IDENTIFICATION

CODE : IF-4-AFQL
ECTS : 1.0

HOURS

Lectures : 9.0 h
Seminars : 8.0 h
Laboratory : 0.0 h
Project : 0.0 h
Teacher-student
contact : 17.0 h
Personal work : 10.0 h
Total : 27.0 h

ASSESSMENT METHOD

DS

TEACHING AIDS

<http://liris.cnrs.fr/pierre-edouard.portier/>

TEACHING LANGUAGE

French

CONTACT

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AIMS

Targeted competence:

- enforce quality assurance and quality control through the derivation of programs correct by construction.

To do this, sub-competences are necessary:

- transform a natural language specification into a formal specification with the predicate calculus
- derive a correct program from its specification

This approach leads to a smart management of the complexity: checking the correction of a program is hard while deriving a program correct by construction splits the complexity in a sequence of simpler decisions.

After this module, one will be able to derive both sequential and concurrent programs correct by construction.

CONTENT

- * Sequential Programs Correct by Construction
- ** Part 1, Theory
- *** Predicate Calculus Reminder
- *** Hoare Triples
- *** Weakest Precondition
- *** Guarded Command Language
- ** Part 2, Examples
- *** Array subsequences [e.g., maximal AB subsequence, longest null subsequence, etc.]
- *** Correct and efficient programs [e.g., integer division, fibonacci, etc.]
- *** Sorting algorithms [Dutch National Flag, Quicksort, etc.]
- * Concurrent Programs Correct by Construction
- ** Part 1, Theory
- *** locally correct / globally correct
- *** System invariant
- *** Weakest liberal precondition
- *** Atomicity
- *** Progress
- ** Partie 2, Examples
- *** Mutual Exclusion of Critical Sections
- *** Safe Sluice
- *** Peterson
- *** Concurrent Linear Search
- *** Election Algorithm
- *** Alternating Bit Protocol

BIBLIOGRAPHY

- * BACKHOUSE, 2002, Program Construction the Correct Way
- * COHEN, 1990, Programming in the 1990s an Introduction to the Calculation of Programs
- * DIJKSTRA, 1976, A Discipline of Programming
- * GRIES, 1981, the Science of Programming
- * KALDEWALL, 1990, Programming the Derivation of Algorithms
- * KOURIE, WATSON, 2012, the Correctness by Construction Approach to Programming

PRE-REQUISITE

IF3-ALGO

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